

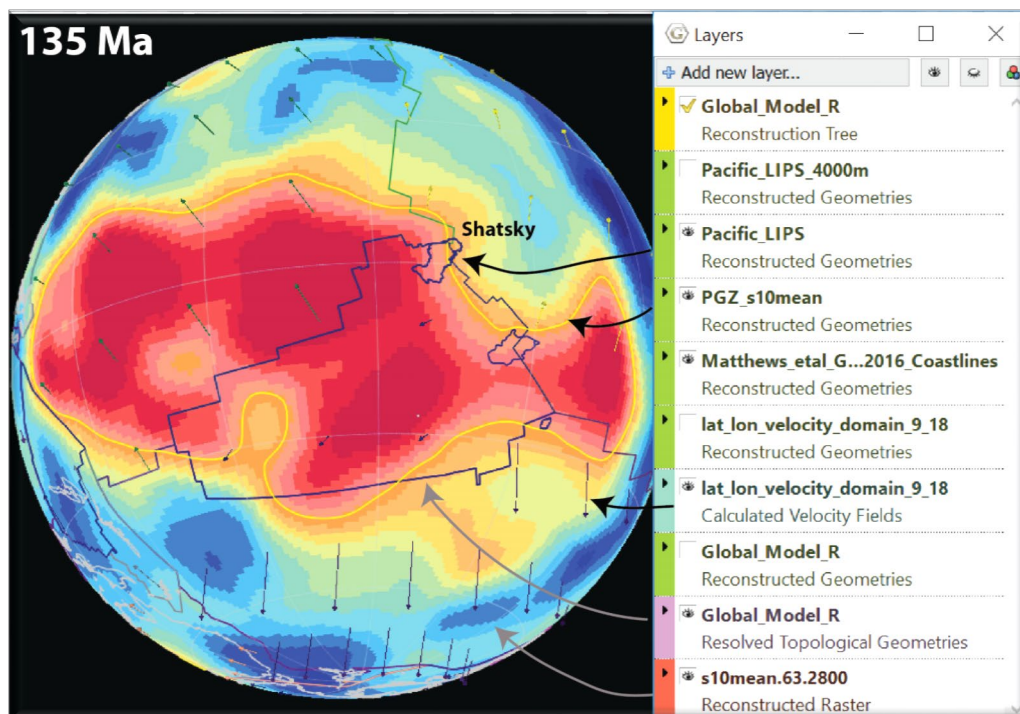
Supplementary files

The following files are included in “PACIFIC.ZIP” which can be uploaded from <http://www.earthdynamics.org/Pacific/PACIFIC.zip>

- Global_Model_R.gproj [GPlates project file that will load all files below]
- Global_Model_R.rot [GPlates rotation file 0-250 Ma]*¹
- Global_Model_R.gplm [GPlates Dynamic plate polygon model]*¹
- Pacific_LIPS.gplm [Pacific LIP polygons used in this paper]
- PGZ_s10mean.gplm [Plume generation zones; *s10mean* 0.9% slow contour]*²
- s10mean.63.2800.gpml [*s10mean* grid model at 2800 km depth]*²
- tomoTHT.cpt [grid palette used with *s10mean* model]

*¹Modified from Matthews et al. (2016)

*²Dobrovine et al. (2016)



Rotation File (Global_Model_R.ROT)

Our plate reconstructions in GPlates format use a hybrid plate motion frame that combines hotspot and true polar wander (TPW) corrected palaeomagnetic (before 125 Ma) reference frames, which can be referred to as a mantle frame (e.g., Dobrovine et al., 2016). The original rotation file is that of Matthews et al. (2016; GPlates 2.0 default) but modifications include (i) revised Pacific extension from 83.5 to 150 Ma (Model R developed here), (ii) critical plate circuits adjusted to conform with GMHRF-3 (Dobrovine et al., 2012) and those used to develop the global apparent polar wander path in

Torsvik et al. (2012), and (iii) included a switch between mantle (Plate Id. = 0) and paleomagnetic (Plate Id. = 1) reference frame.

Adjusted rotations are marked with “CEED” in the right-hand comment/information column, the rotation file works back to 250 Ma, and can be used to reconstruct all other files supplied with GPlates 2.0/2.1 (e.g., magnetic isochrones, coastlines). The rotation file does not work satisfactory with continents and terranes published in Torsvik & Cocks (2017) due to some differing plate numbers.

At start-up, the GPlates software defaults to a mantle frame (Plate Id = 0). In order to show our reconstructions with respect to the spin-axis (palaeomagnetic or paleogeographic frame), select option “Reconstruction”, sub-option “Specify Anchored Plate ID”, and type 1.

Plate reconstructions that aims to explore paleoclimate (e.g. plotting latitudinal sensitive sedimentary facies) or comparing reconstructions with global climate simulations or other paleomagnetic reconstructions should always be reconstructed with Plate Id = 1. Conversely, Plate Id = 0 (mantle frame) should be used as input for mantle modelling and for example to compare surface volcanism such as large igneous provinces with deep mantle mantle anomalies.

Dynamic Plate Polygon Model (Global_Model_R.gplm)

GPlates allows full-plate model construction using time-dependent plate boundaries which self-close at any arbitrary spatial or temporal scale to form dynamic polygons (Gurnis et al., 2012). The original model is that of Seton et al. (2012) developed back to 200 Ma. Here we use the model of Matthews et al. (2016) back to 250 Ma (Default GPlates 2.0) but because we have revised critical plate circuits and rotation poles (including a new Model R for the Pacific), the polygon model required many modifications to close the polygons appropriately; we have not added new plates or done radical changes in the original plate polygon model.